Mechanics of Human Movement

Course Progress 90.2% (11 of 13)

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Assessment

Week 3: Assignment 3

Due date on 2026-08-14, 23:59 IST

The due date for submitting this assignment has passed.Your last recorded submission was on 2026-08-08, 22:16 IST.

1. Force generation is reduced at high velocities of muscle contraction because of 1 Point

○ Higher number of sarcomere units at higher velocities
○ Thicker sarcolemma at higher velocities
● Fewer cross bridges formed at higher velocities
○ More straight bridges formed at higher velocities

Yes, the answer is correct. Score: 1

Accepted answer is Fewer cross bridges formed at higher velocities

2. A batsman has to flex her elbow while playing a cricket shot. The biceps brachii helps in flexing the elbow and has points of origin and insertion at scapula and radius respectively. Which kind of muscle will help in keeping the scapula intact to help the batsman complete her shot successfully? 1 Point

○ Agonist
○ Antagonist
○ Neutralizer
● Stabilizer

Yes, the answer is correct. Score: 1

Accepted answer is Stabilizer

3. When a person flexes her elbow while carrying a dumbbell in her hand, the biceps brachii will be in 1 Point

● Concentric Action
○ Eccentric Action
○ Isometric Action
○ Esoteric Action

Yes, the answer is correct. Score: 1

Accepted answer is Concentric Action

4. Passive tension and viscoelastic nature play a greater role in 1 Point

○ Concentric Action
● Eccentric Action
○ Isometric Action
○ Isoteric Action

Yes, the answer is correct. Score: 1

Accepted answer is Eccentric Action

5. Cross-bridges formed in the sarcomere of a muscle are highest in 1 Point

○ Concentric Action
● Eccentric Action
○ Isometric Action
○ Isoteric Action

No, the answer is incorrect. Score: 0

Accepted answer is Isometric Action

6. As a mechanical equivalent, a muscle can be assumed to be a 1 Point

○ Single force member
● Two force member
○ Three force member
○ Four force member

Yes, the answer is correct. Score: 1

Accepted answer is Two force member

7. Which of the following muscles cause elbow flexion? 1 Point

☑ Biceps brachii
☑ Brachialis
☑ Brachioradialis
☐ Biceps Femoris

Yes, the answer is correct. Score: 1

Accepted answer is Biceps brachii, Brachialis, Brachioradialis

8. Elbow joint is a 1 Point

☐ Cartilaginous joint
☑ Synovial joint
☐ Fibrous joint
☑ Diarthrodial joint

Yes, the answer is correct. Score: 1

Accepted answer is Synovial joint, Diarthrodial joint

9. What type of lever is a bent arm? 1 Point

○ Class I lever
○ Class II lever
● Class III lever
○ Class IV lever

Yes, the answer is correct. Score: 1

Accepted answer is Class III lever

10. While drawing a free body diagram, we include 1 Point

☑ External forces on the body
☐ Internal forces of the system that are external to the body
☐ Internal forces on the body
☐ Internal moments on the body

Partially Correct. Score: 0.5

Accepted answer is External forces on the body, Internal forces of the system that are external to the body

11. Which of the following strategies can be adopted for solving a statically indeterminate system in biomechanics? 1 Point

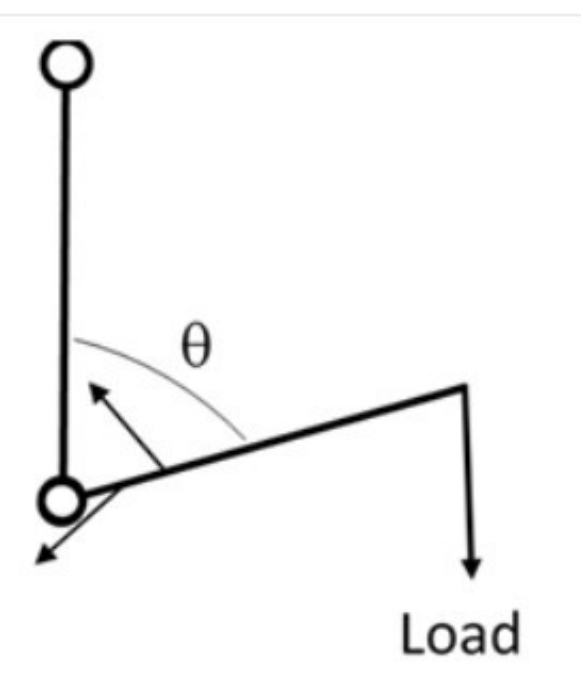
☑ Relationship between muscle force and the cross sectional area of the muscle
☑ Optimization to reduce metabolic cost, joint reaction etc.
☑ Relationship between muscle force and EMG signal
☑ Relationship using velocities and accelerations of individual segments

No, the answer is incorrect. Score: 0

Accepted answer is Relationship between muscle force and the cross sectional area of the muscle, Optimization to reduce metabolic cost, joint reaction etc., Relationship between muscle force and EMG signal

12. Consider a 160cm tall boy (H=1.6m) weighing 60kg holding a mass of 5kg in his hand as shown in the figure below. The forearm make an angle θ with the upper arm, which is vertical. The insertion points of the biceps brachii and triceps brachii, are 0.1m and 0.05m from the elbow, respectively. Their lines of action are at 30 degrees and 10 degrees, respectively, to the axis of the forearm as shown. Assuming the arm is in static equilibrium at θ = 90°, answer Q12-15. Use appropriate anthropometric data provided for calculating the length, weight and the distance of the centre of mass (CoM) from the proximal joint of the individual segments. Assume g=10 m/s^2. 1 Point

Given:



$L_{forearm}$ = 0.2H
$W_{forearm}$ = 2%W
$CoM_{forearm} = 0.5L_{forearm}$

Choose the answer closest to the solution you have obtained.

The moment about the elbow joint due to the external forces is

○ 2 Nm
○ 8 Nm
○ 16 Nm
● 18 Nm

Yes, the answer is correct. Score: 1

Accepted answer is 18 Nm

13. Assuming only the biceps brachii is acting, calculate the force exerted by it. 1 Point

○ 704N
● 358N
○ 70.4N
○ 60N

Yes, the answer is correct. Score: 1

Accepted answer is 358N

14. If the triceps are also acting, the force in the biceps muscle 1 Point

○ Will be the same as in Q13
● Will be greater than the value in Q13
○ Will be lesser than the value in Q13
○ Will be independent of the load being carried

Yes, the answer is correct. Score: 1

Accepted answer is Will be greater than the value in Q13

15. Assume that the ratio of cross-sectional area of the biceps brachii to the triceps is 2. The force in the triceps muscle is 1 Point

○ 98N
● 196N
○ 392N
○ 784N

Yes, the answer is correct. Score: 1

Accepted answer is 196N